

Classification problem

Overfitting and regularization

Khiem Nguyen

Email	khiem.nguyen@glasgow.ac.uk
MS Teams	khiem.nguyen@glasgow.ac.uk
Whatsapp	+44 7729 532071 (Emergency only)

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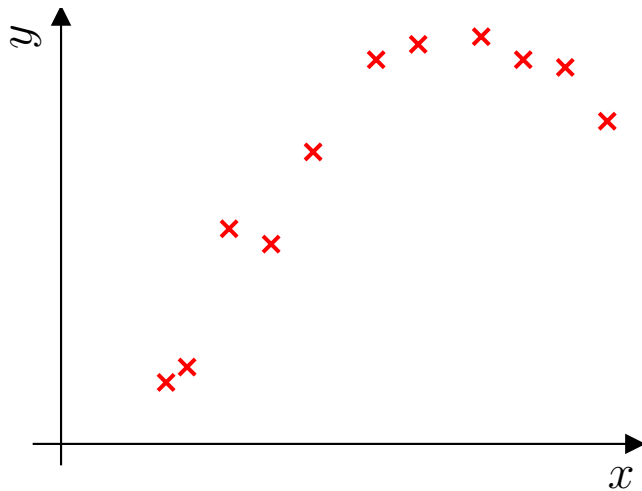


University
of Glasgow

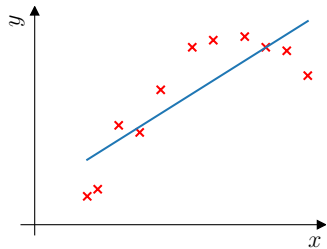
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- 2 Addressing overfitting
- 3 Regularization: Intuition and Formulation

Underfitting versus overfitting



Underfit versus overfit



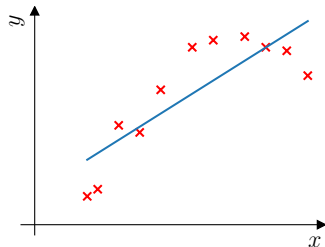
Underfitting

$$w_1x + b$$

Does not fit the training set
well

high bias

Underfit versus overfit

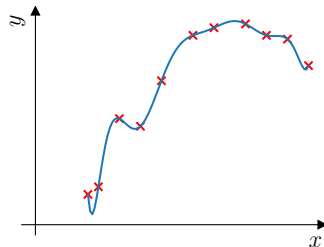


Underfitting

$$w_1x + b$$

Does not fit the training set well

high bias



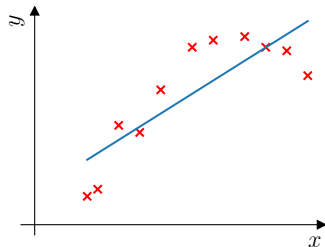
Overfitting

$$w_1x + w_2x^2 + \dots + w_{10}x^{10} + b$$

Fit the training set extremely well

high variance

Underfit versus overfit

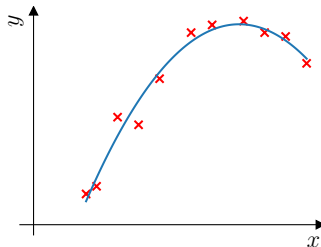


Underfitting

$$w_1x + b$$

Does not fit the training set well

high bias

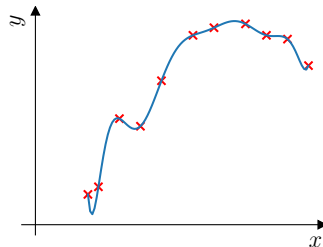


Just right

$$w_1x + w_2x^2 + b$$

Fit training set pretty well

generalization



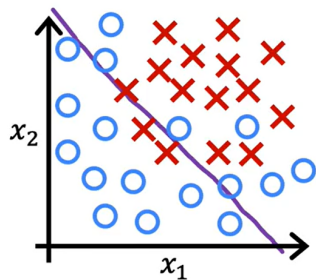
Overfitting

$$w_1x + w_2x^2 + \dots + w_{10}x^{10} + b$$

Fit the training set extremely well

high variance

Underfitting versus overfitting: Classification



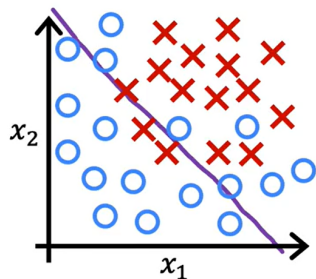
Underfit – high bias

$$z = w_1x_1 + w_2x_2 + b$$

$$f_{\vec{w},b}(\vec{x}) = g(z)$$

g is the sigmoid function

Underfitting versus overfitting: Classification

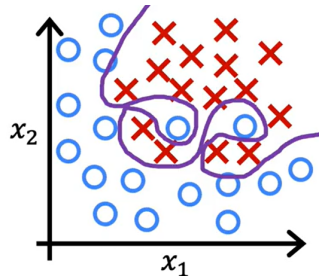


Underfit – high bias

$$z = w_1x_1 + w_2x_2 + b$$

$$f_{\vec{w},b}(\vec{x}) = g(z)$$

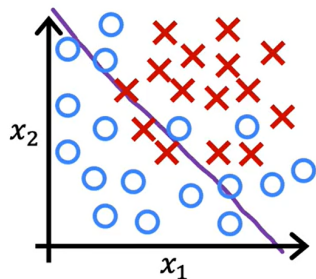
g is the sigmoid function



Overfit - high variance

$$\begin{aligned} z = & w_1x_1 + w_2x_2 + w_3x_1^2x_2 \\ & + w_4x_1^2x_2^2 + w_5x_1^2x_2^3 \\ & + w_6x_1^3x_2 + \dots + b \end{aligned}$$

Underfitting versus overfitting: Classification

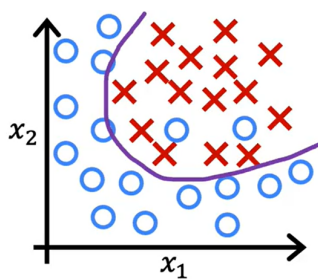


Underfit – high bias

$$z = w_1x_1 + w_2x_2 + b$$

$$f_{\vec{w},b}(\vec{x}) = g(z)$$

g is the sigmoid function

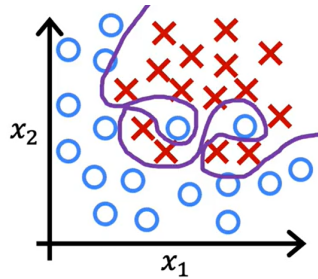


Just right

$$z = w_1x_1 + w_2x_2$$

$$+ w_3x_1^2 + w_4x_2^2$$

$$+ w_5x_1x_2 + b$$



Overfit - high variance

$$z = w_1x_1 + w_2x_2 + w_3x_1^2x_2$$

$$+ w_4x_1^2x_2^2 + w_5x_1^2x_2^3$$

$$+ w_6x_1^3x_2 + \dots + b$$

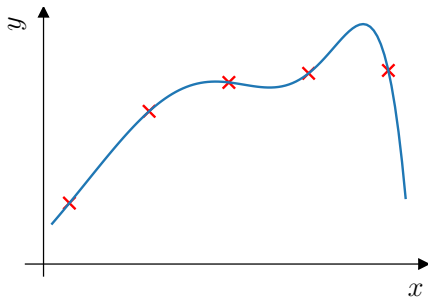
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① Underfitting (high bias) versus Overfitting (high variance)

② Addressing overfitting

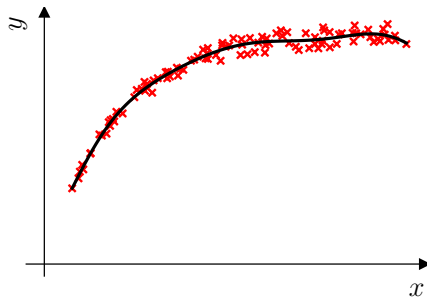
③ Regularization: Intuition and Formulation

Addressing overfitting: Collect more training examples



Overfit

Maybe reduce the order of fitting?



Collect more training examples

It is not always easier to harvest more data

Select features to include/exclude

size	bedrooms	floors	age	avg. income	...	distance to center	price
x_1	x_2	x_3	x_4	x_5		x_{100}	y

Select features to include/exclude

size	bedrooms	floors	age	avg. income	...	distance to center	price
x_1	x_2	x_3	x_4	x_5		x_{100}	y

all features
+
insufficient data
↓
overfit

Select features to include/exclude

size	bedrooms	floors	age	avg. income	...	distance to center	price
x_1	x_2	x_3	x_4	x_5		x_{100}	y

all features
+
insufficient data
↓
overfit

selected features

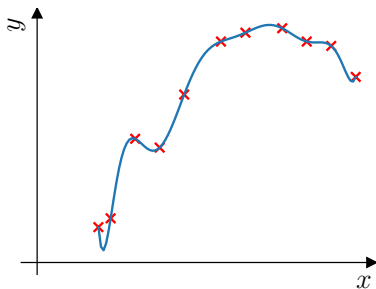
size
bedrooms
age
distance to center

just right feature selection

disadvantage

↓
useful features
could be lost

Regularization

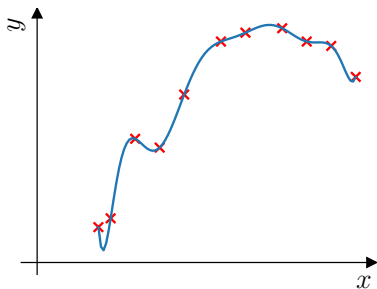


$$f_{\vec{w},b}(\vec{x}) =$$
$$-8246.12x + 0.1351x^2 - \dots + 33781x^8 - 542x^9 +$$
$$33.92x^{10} + 974.89$$

large values for model parameters $\vec{w}, b$

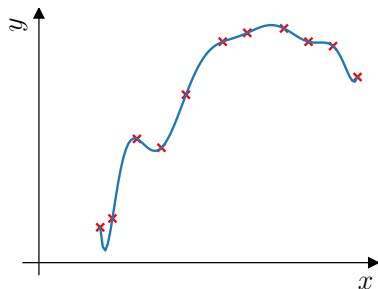
Produced by using hand-written code with regularization on $w_3, \dots, w_{10}$

Regularization



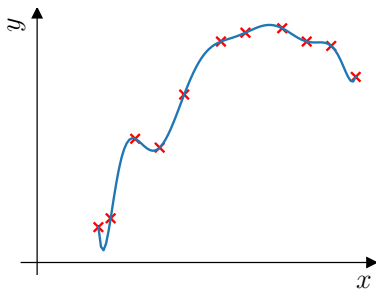
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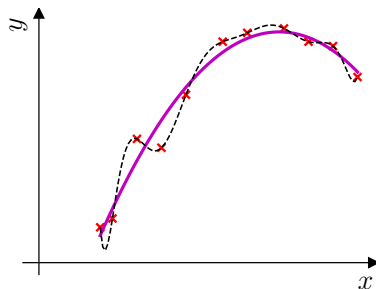
Produced by using hand-written code with regularization on $w_3, \dots, w_{10}$

Regularization



$$f_{\vec{w},b}(\vec{x}) = -8246.12x + 0.1351x^2 - \dots + 33781x^8 - 542x^9 + 33.92x^{10} + 974.89$$

large values for model parameters $\vec{w}, b$



$$f_{\vec{w},b}(\vec{x}) = 2.971x_{\text{scaled}} - 2.35x_{\text{scaled}}^2 + \text{small-no.}x_{\text{scaled}}^3 + \dots + \text{small-no.}x_{\text{scaled}}^4 + \text{small-no.}x_{\text{scaled}}^{10} + 1.845$$

smaller values for model parameters $\vec{w}, b$

Produced by using hand-written code with regularization on $w_3, \dots, w_{10}$

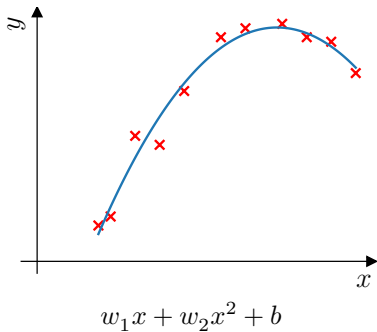
Options

- ① Collect more data
- ② Select features (feature selection)
- ③ Reduce size of parameters (**regularization!**) → We study it now!

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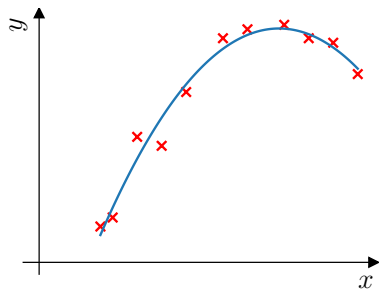
- ① Underfitting (high bias) versus Overfitting (high variance)
- ② Addressing overfitting
- ③ Regularization: Intuition and Formulation

Regularization: Intuition



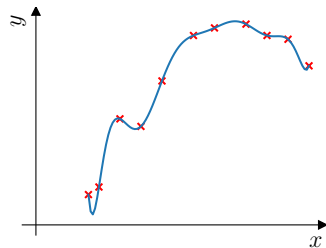
Idea: Make $w_3, w_4, \dots, w_{10}$ really small (≈ 0)

Regularization: Intuition



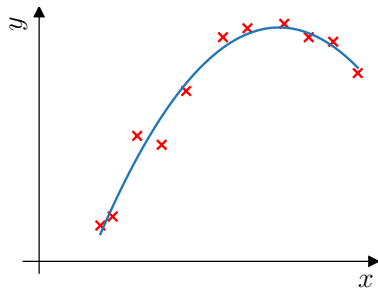
$$w_1x + w_2x^2 + b$$

Idea: Make $w_3, w_4, \dots, w_{10}$ really small (≈ 0)



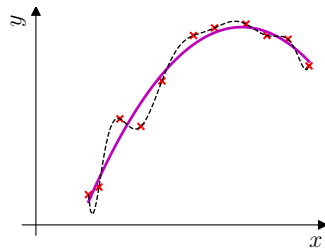
$$w_1x + w_2x^2 + w_3 \times x^3 + \dots + w_{10} \times x^{10} + b$$

Regularization: Intuition



$$w_1x + w_2x^2 + b$$

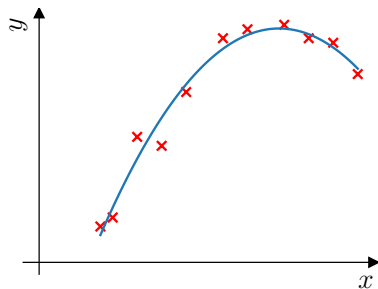
Idea: Make $w_3, w_4, \dots, w_{10}$ really small (≈ 0)



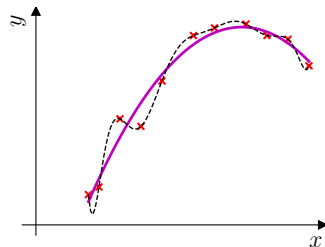
$$w_1x + w_2x^2$$

$$+ \cancel{w_3} \overset{\approx 0}{\times} x^3 + \dots + \cancel{w_{10}} \overset{\approx 0}{\times} x^{10} + b$$

Regularization: Intuition



$$w_1x + w_2x^2 + b$$



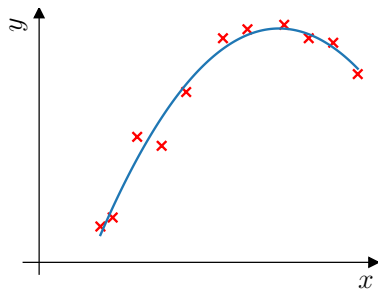
$$w_1x + w_2x^2$$

$$+ \cancel{w_3} \overset{\approx 0}{\times} x^3 + \dots + \cancel{w_{10}} \overset{\approx 0}{\times} x^{10} + b$$

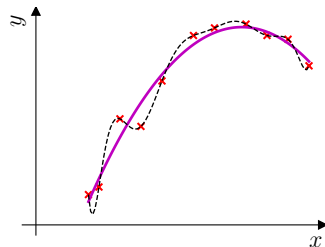
Idea: Make $w_3, w_4, \dots, w_{10}$ really small (≈ 0)

$$\min_{\vec{w}, b} \left\{ \frac{1}{2m} \sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)})^2 \right\}$$

Regularization: Intuition



$$w_1x + w_2x^2 + b$$



$$w_1x + w_2x^2$$

$$+ \cancel{w_3} \overset{\approx 0}{\times} x^3 + \dots + \cancel{w_{10}} \overset{\approx 0}{\times} x^{10} + b$$

Idea: Make $w_3, w_4, \dots, w_{10}$ really small (≈ 0)

$$\min_{\bar{w}, b} \frac{1}{2m} \sum_{i=1}^m (f_{\bar{w}, b}(\vec{x}^{(i)}) - y^{(i)})^2 + \text{Large-number} \times w_3^2 + \dots + \text{Large-number} \times w_{10}^2$$

To make the cost function small, $w_3, \dots, w_{10}$ should become smaller ($\rightarrow 0$).

Regularization: Cost function

size	bedrooms	floors	age	avg. income	...	distance to center	price
x_1	x_2	x_3	x_4	x_5		x_{100}	y

Smaller values $w_1, w_2, \dots, w_n, b \rightarrow$ simpler model $\rightarrow$ less likely to overfit

$$J(\vec{w}, b) = \frac{1}{2m} \sum_{i=1}^m \left(f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)} \right)^2 + \underbrace{\frac{\lambda}{2m} \sum_{j=1}^n w_j^2}_{\text{regularization term}}$$

λ – **regularization parameter**

Remember: $\lambda > 0$

Regularization

$$\min_{\vec{w}, b} J(\vec{w}, b) = \min_{\vec{w}, b} \left\{ \underbrace{\frac{1}{2} \frac{1}{m} \sum_{i=1}^m \left(f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)} \right)^2}_{\text{mean squared error}} + \underbrace{\frac{\lambda}{2m} \sum_{j=1}^n w_j^2}_{\text{regularization term}} \right\}$$

λ balances both goals

- ① fit the data
- ② keep w_j small (if needed)

But

choosing large λ may lead to simple fit. For example, with $\lambda = 10^{10}$

$$f_{\vec{w}, b}(\vec{x}) = \cancel{w_1} \approx 0 x + \cancel{w_2} \approx 0 x^2 + \dots + \cancel{w_{10}} \approx 0 x^{10} + b \rightarrow f_{\vec{w}, b}(\vec{x}) = 0$$

Regularized linear regression

$$\min_{\vec{w}, b} J(\vec{w}, b) = \min_{\vec{w}, b} \left\{ \underbrace{\frac{1}{2m} \sum_{i=1}^m \left(f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)} \right)^2}_{1/2 \times \text{MSE}} + \underbrace{\frac{\lambda}{2m} \sum_{j=1}^n w_j^2}_{\text{regularization term}} \right\}$$

Gradient descent

Repeat

$$\begin{cases} w_1 = w_1 - \alpha \frac{\partial J}{\partial w_1}(\vec{w}, b) \\ \vdots \\ w_n = w_n - \alpha \frac{\partial J}{\partial w_n}(\vec{w}, b) \\ b = b - \alpha \frac{\partial J}{\partial b}(\vec{w}, b) \end{cases}$$

$$\frac{\partial J}{\partial w_j} = \underbrace{\frac{1}{m} \sum_{i=1}^m \left(f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)} \right) x_j^{(i)}}_{\text{from } 1/2 \times \text{MSE}} + \frac{\lambda}{m} w_j \quad j = 1, \dots, n$$

$$\frac{\partial J}{\partial b} = \frac{1}{m} \sum_{i=1}^m \left(f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)} \right)$$

Don't have to regularize b

Implementing gradient descent

Repeat

$$\begin{cases} w_j = w_j - \alpha \left\{ \frac{1}{m} \sum_{i=1}^m [(f_{\vec{w},b}(x^{(i)}) - y^{(i)}) x_j^{(i)}] + \frac{\lambda}{m} w_j \right\} \\ b = b - \alpha \frac{1}{m} \sum_{i=1}^m (f_{\vec{w},b}(x^{(i)}) - y^{(i)}) \end{cases}$$

$$w_j = \underbrace{w_j - \alpha \frac{\lambda}{m} w_j}_{w_j \left(1 - \alpha \frac{\lambda}{m} \right)} - \underbrace{\alpha \frac{1}{m} \sum_{i=1}^m (f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)}}_{\text{usual update as without regularization}}$$

The term $w_j \left(1 - \alpha \frac{\lambda}{m} \right)$ **shrinks** w_j a bit in each iteration

Example:

$$\alpha \frac{\lambda}{m} = 0.1 \times \frac{1}{1000} = 0.0001$$

$$\rightarrow w_j \underbrace{(1 - 0.0001)}_{0.9999}$$

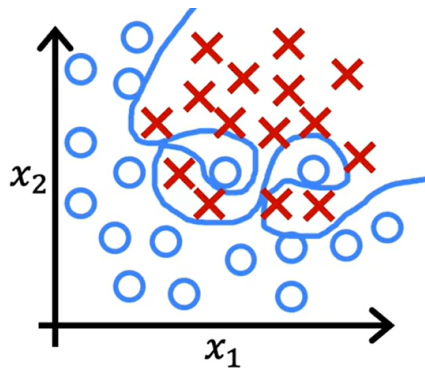
Derivation of gradient of the cost function $J(\vec{w}, b)$ (optional)

$$\begin{aligned}\frac{\partial J}{\partial w_1} J(\vec{w}, b) &= \frac{\partial}{\partial w_1} \left\{ \frac{1}{2m} \sum_{i=1}^m \left(\underbrace{\vec{w} \cdot \vec{x}^{(i)} + b - y^{(i)}}_{w_1 x_1^{(i)} + \dots + w_n x_n^{(i)} + b - y^{(i)}} \right)^2 + \frac{\lambda}{2m} \sum_{j=1}^n w_j^2 \right\} \\&= \frac{1}{2m} \sum_{i=1}^m 2 \left[\left(\vec{w} \cdot \vec{x}^{(i)} + b - y^{(i)} \right) x_1^{(i)} \right] + \frac{\lambda}{2m} 2w_1 \\&= \frac{1}{m} \sum_{i=1}^m \left[\left(\underbrace{\vec{w} \cdot \vec{x}^{(i)} + b - y^{(i)}}_{f_{\vec{w}, b}(\vec{x}^{(i)})} \right) x_1^{(i)} \right] + \frac{\lambda}{m} w_1 = \frac{1}{m} \sum_{i=1}^m \left[\left(f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)} \right) x_1^{(i)} \right] + \frac{\lambda}{m} w_1\end{aligned}$$

Generalization ($\partial J / \partial b$ the same as before):

$$\frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_{i=1}^m \left[\left(f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)} \right) x_j^{(i)} \right] + \frac{\lambda}{m} w_j$$

Regularized logistic regression

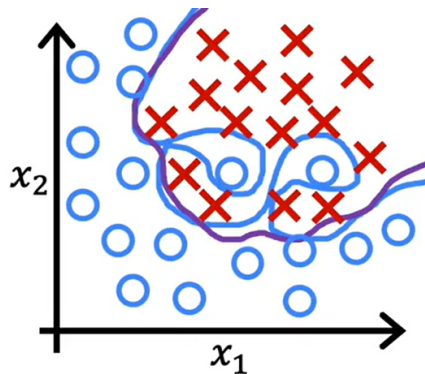


$$z = w_1x_1 + w_2x_2 + w_3x_1^2x_2$$

$$+ w_4x_1^2x_2^2 + w_5x_1^2x_2^3 + \dots + b$$

$$f_{\bar{w},b} = \frac{1}{1 + e^{-z}}$$

Regularized logistic regression



$$z = w_1 x_1 + w_2 x_2 + w_3 x_1^2 x_2$$

$$+ w_4 x_1^2 x_2^2 + w_5 x_1^2 x_2^3 + \dots + b$$

$$f_{\vec{w}, b} = \frac{1}{1 + e^{-z}}$$

Cost function

$$J(\vec{w}, b) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log \left(f_{\vec{w}, b}(\vec{x}^{(i)}) \right) + (1 - y^{(i)}) \log \left(1 - f_{\vec{w}, b}(\vec{x}^{(i)}) \right) \right] + \frac{\lambda}{2m} \sum_{j=1}^n w_j^2$$

$$\min_{\vec{w}, b} J(\vec{w}, b) \quad \rightarrow \quad w_j \downarrow$$

Regularized logistic regression:

$$J(\vec{w}, b) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log \left(f_{\vec{w}, b}(\vec{x}^{(i)}) \right) + (1 - y^{(i)}) \log \left(1 - f_{\vec{w}, b}(\vec{x}^{(i)}) \right) \right] + \frac{\lambda}{2m} \sum_{j=1}^n w_j^2$$

Gradient descent

Repeat

$$\begin{cases} w_j = w_j - \alpha \frac{\partial J}{\partial w_j}(\vec{w}, b) \\ j = 1, \dots, n \\ b = b - \alpha \frac{\partial J}{\partial b}(\vec{w}, b) \end{cases}$$

$$\frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_{i=1}^m \left(f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)} \right) x_j^{(i)} + \frac{\lambda}{m} w_j$$

$$j = 1, \dots, n$$

$$\frac{\partial J}{\partial b} = \frac{1}{m} \sum_{i=1}^m \left(f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)} \right)$$

Looks same as for linear regression! **BUT**

$f_{\vec{w}, b}(\vec{x}) = g(z(\vec{x}; \vec{w}, b))$ is logistic regression function model